AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 31 · STUDENT LESSON

Design the Decision Process

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Validate. Check both conditions. Advance—or stop.

Continue the activity project with visible candidate state, complete branches and a reason the search terminates.

Keep one habit: Keep the request fixed → advance after rejection → stop at the first fit or end.

MAIN LESSON · SECTION 1

Where does the process go next—and when does it stop?

> Preserve the contract while making every branch traceable.

Start with the missing branch

A complete flow must say what happens when the first row fails, when no row fits and when the request is unusable. Each search step needs visible progress toward a stopping point.

Recall missing versus valid zero

For the activity project, how does a blank time differ from a valid request for zero minutes?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your **explanation** that the **evidence** shows was wrong.

Your learning goals

- I can design a decision **process** that covers valid, invalid and no-match requests.
- I can trace changing **state** and explain why the search stops.
- I can preserve both compatibility conditions and the agreed first-match rule.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

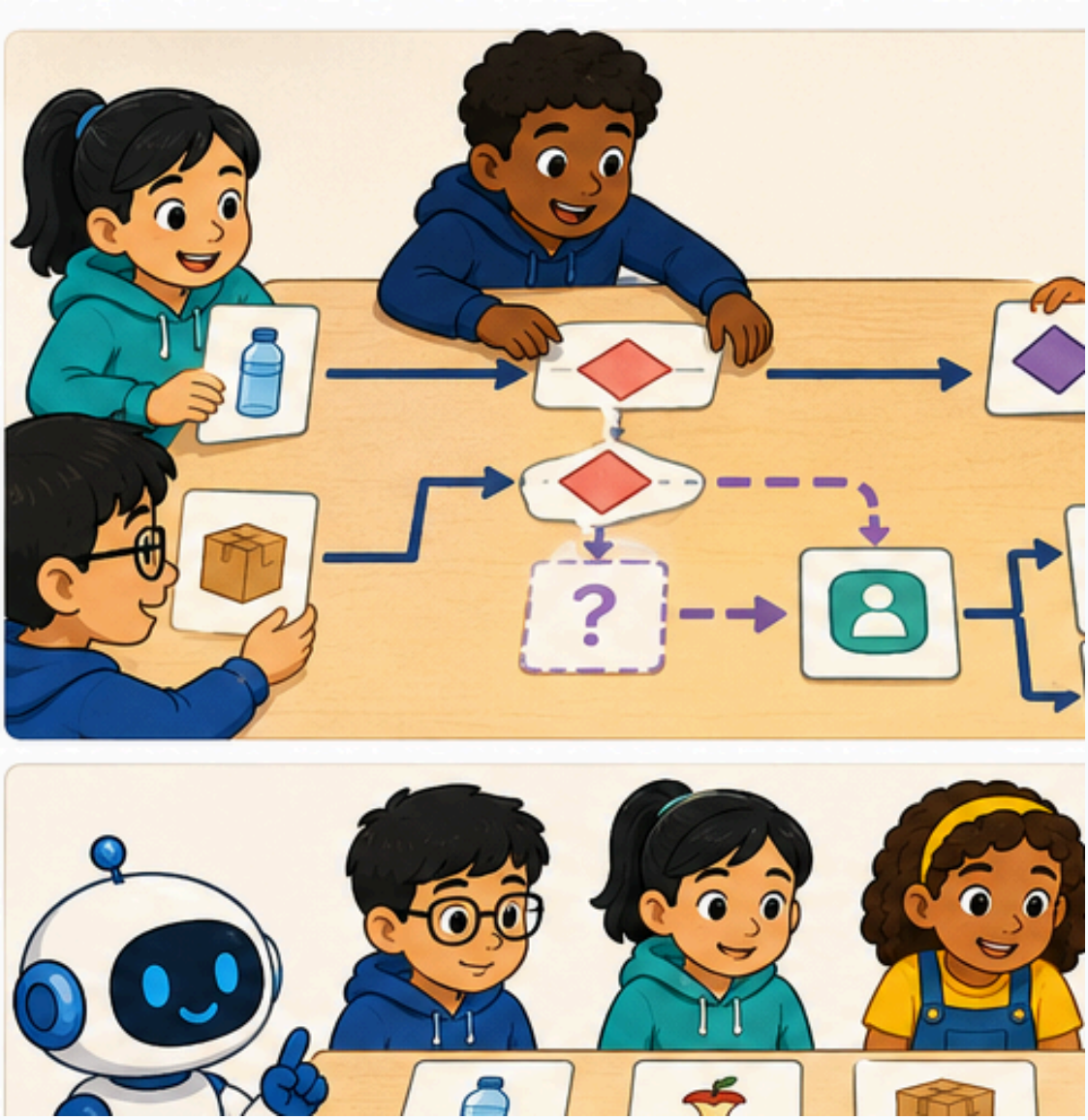
Remember before reading

Without opening the old lesson, explain one key idea from Choose **Input**s and **Output**s. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Humans Set Boundaries. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Generate, Retrieve, or Calculate?. Give an example and a mistake that the idea helps you avoid.

Rules or learned patterns must connect **input**s to **output**s transparently.
</details>



LEARNING TARGET Rule: if paper then P, otherwise M. **Input**s are paper, metal, glass and unknown. Repair the rule for paper and metal trays without inventing a glass tray. Trace all four **input**s.

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can design a complete **decision flow** with **validation**, search and stopping.

MAIN LESSON · SECTION 2

Carry forward the activity contract unchanged

> Continue the same B1–B4 activity project from Week 30.

Define a complete process

A complete **process** says what happens for every kind of request in the task. It must handle a **fitting** activity, no **fitting** activity and unusable **input**. We will design a paper algorithm before building the prototype.

Its decisions use the requirements from Week 30, so a different drawing should not silently change what counts as correct.

Read the catalogue and input rules

Keep the same catalogue order: B1 Bookmark drawing, 10 minutes, quiet Yes; B2 Story circle, 8 minutes, quiet No; B3 Pattern puzzle, **12 minutes**, quiet Yes; B4 Tower challenge, 15 minutes, quiet No.

These are fixed paper-model values with materials assumed available. Valid **input** is a whole number of minutes from 0 to 30 and Quiet required Yes or No.

Require both conditions and the first fit

A **candidate** fits when its required minutes are no more than the available minutes and either quiet is not required or the activity is quiet. Both parts must hold.

The **output** is the first **fitting** activity in catalogue order, with a reason, or a clear no-match result. Invalid or missing **input** produces a clarification request. None of these **output**s books materials or publishes a message.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A transparent sorter first checks item type, then one disambiguating feature, then sends unclear items to review.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and **state** the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

MAIN LESSON · SECTION 3

Design validation, search and stopping

> A corrected request starts a fresh run with fresh candidate state.

Inspect the four branch outcomes

Four paths through the recommender

CONDITION REACHED	NEXT ACTION	WHERE THE PATH ENDS
Input missing or invalid	Ask for corrected input	Stop this run
Candidate fails a check	Move to the next row	Continue search
Candidate passes both	Suggest it with a reason	Stop this run
All rows checked and failed	Report no match	Stop this run

Failing one candidate continues the search. No match is justified only after the valid request has no fitting candidate.

Validate all request fields before searching

Begin by checking the request. If minutes are missing, outside 0 to 30 or not a whole number, ask for a whole number in that range. If the quiet field is not Yes or No, ask for one of those values.

State the fields needing correction and stop this run. When a corrected request arrives, validate it as a new run rather than continuing with old **candidate state**.

Advance one row or stop with a justified result

- For a valid request, start at B1. Check time and the quiet condition for the current **candidate**. If both pass, **output** that **candidate** and stop. If either fails, move forward exactly one row.

  After B4 fails, there is no next row, so **output** no match and stop. Never jump back to B1 within the same search.

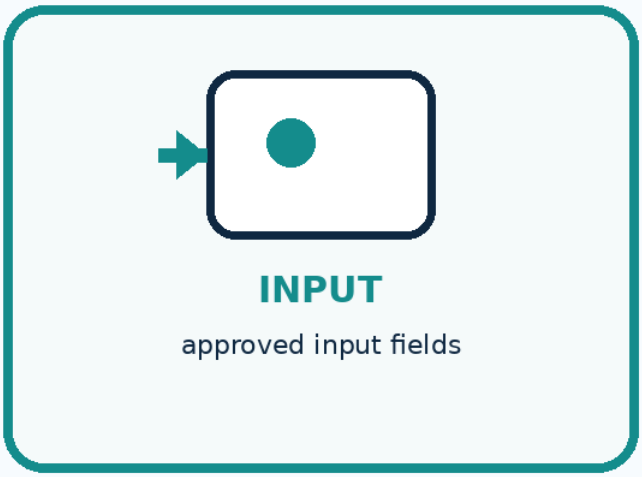
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

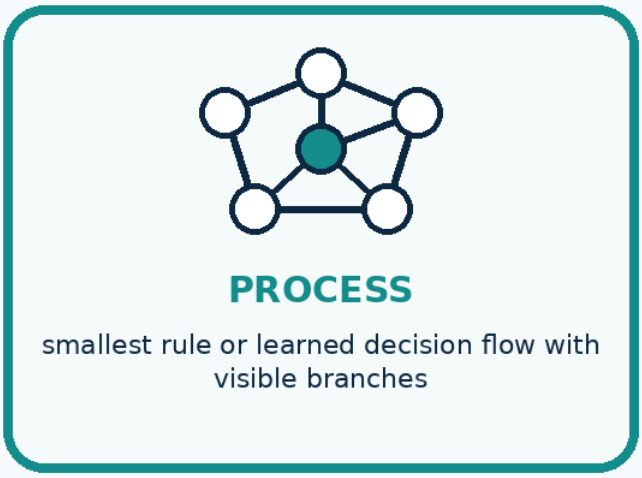
▼ Earlier wording, alternatives and picture transcripts

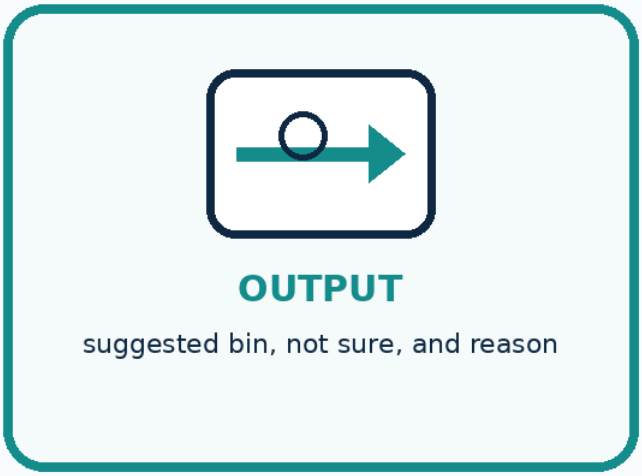
VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT approved **input** fields

PROCESS smallest rule or learned **decision flow** with visible **branch**es

OUTPUT suggested bin, not sure, and reason

CORE IDEA Rules or learned patterns must connect **input**s to **output**s transparently.

UNDER THE HOOD A decision **process** operationalizes the interface contract. Simplicity improves testability and **explanation**, but oversimplification can erase relevant distinctions.

Learned components need explicit data, threshold, and fallback documentation.

Draw a fresh example with all three stages:

A closer explanation

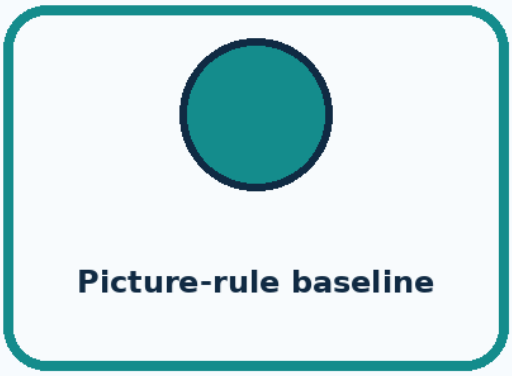
Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare visible decision paths and incomplete fallbacks

> An inspectable flow still needs evidence that its decisions meet the requirements.

Pictures paired with their evidence



The case to investigate

CASE 1 — Picture-rule **baseline** Clue: direct visible **branch**es. Meaning: easy to inspect.



The case to investigate

CASE 2 — Learned classifier Clue: patterns from examples. Meaning: needs data **evidence**.



The case to investigate

CASE 3 — Extra unused field Clue: never changes path. Meaning: remove.



The case to investigate

CASE 4 — No fallback Clue: forces ambiguous label. Meaning: add review.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

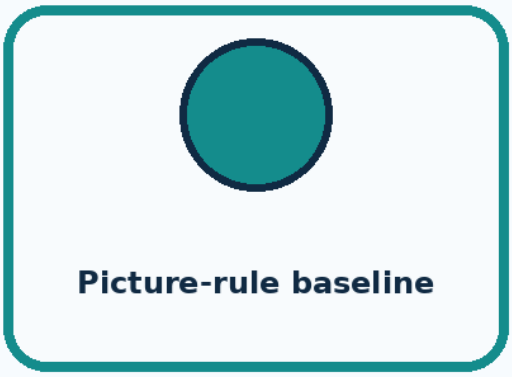
BOUNDARY Transparent does not mean correct; the flow still needs testing.

▼ Earlier wording, alternatives and picture transcripts

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Picture-rule **baseline** Clue: direct visible **branch**es. Meaning: easy to inspect.

CASE 2 — Learned classifier Clue: patterns from examples. Meaning: needs data **evidence**.

CASE 3 — Extra unused field Clue: never changes path. Meaning: remove.

CASE 4 — No fallback Clue: forces ambiguous label. Meaning: add review.

Change one thing in the pictured case

Change: Increase scissors needed to 3.

Prediction: K3 is selected.

Reason: K1 lacks glue; K2 has only 2 scissors; K3 has glue and 4.

Trace the separate craft-kit decision flow

> K1–K3 are kit records, separate from the activity catalogue.

Read the kit records and requested conditions

A fictional worksheet routes craft kits by required glue and number of scissors.

K1: glue No, scissors 2. K2: glue Yes, scissors 2. K3: glue Yes, scissors 4.

Request: glue required Yes, scissors needed 2. Return first kit meeting both requirements.

Compare with the activity recommender's four paths

AI ACADEMY / AI-1/31 · 03 · INSPECT THE LESSON MODEL

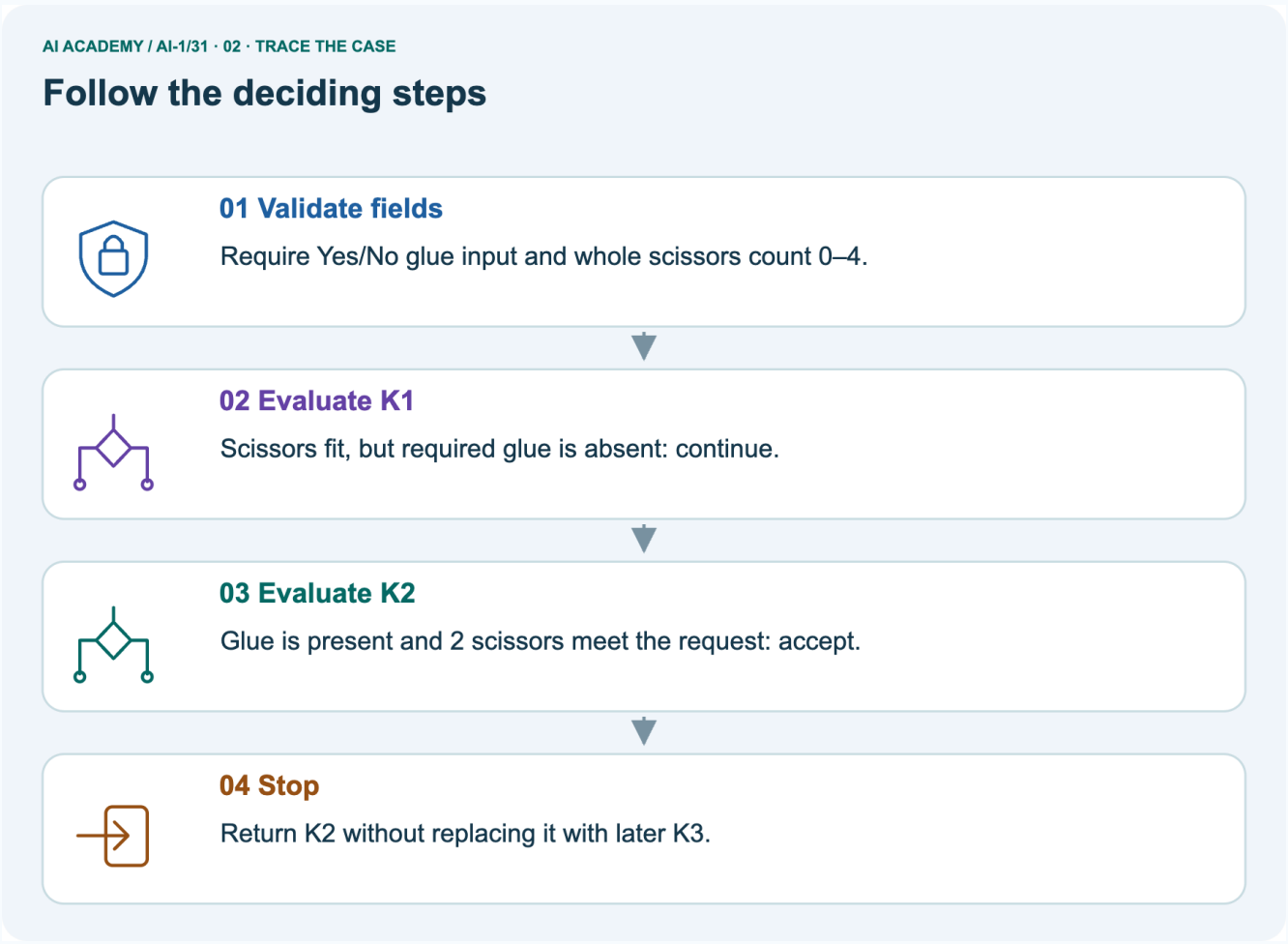
Four paths through the recommender

Condition reached	Next action	Where the path ends
Input missing or invalid	Ask for corrected input	Stop this run
Candidate fails a check	Move to the next row	Continue search
Candidate passes both	Suggest it with a reason	Stop this run
All rows checked and failed	Report no match	Stop this run

Failing one candidate continues the search. No match is justified only after the valid request has no fitting candidate.

Follow the labelled picture with the matching explanation. Open the vector diagram

Trace both checks and the first-match stop



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Validate fields	Require Yes/No glue input and whole scissors count 0–4.
2. Evaluate K1	Scissors fit, but required glue is absent: continue.
3. Evaluate K2	Glue is present and 2 scissors meet the request: accept.
4. Stop	Return K2 without replacing it with later K3.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Explain the rejected row and unvisited later row

- A complete decision **process** specifies what to do after each result. One passing requirement is insufficient.

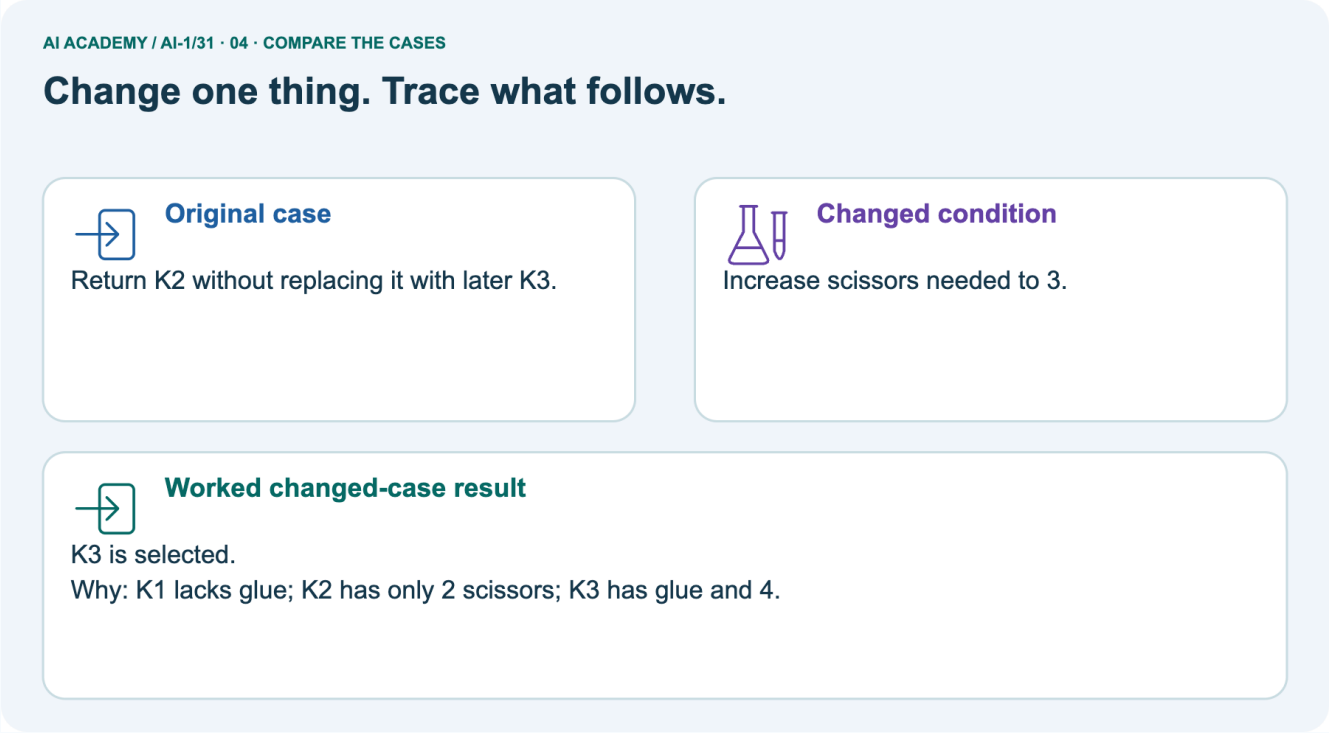
  After a kit passes both, stopping preserves the first-match contract; automatically choosing the largest later kit would implement a different rule. A full scan with no accepted kit returns no match.

- Read the labelled picture: Where does the first original rejection occur, and why is K3 not originally selected?

Predict the three-scissors request

- Before opening the changed-scissors picture, keep glue required Yes and increase the request from two scissors to three. Trace K1, K2 and K3 in order. Which failed condition makes each rejected kit advance to the next row?

Check the changed kit trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the inventory assumption visible

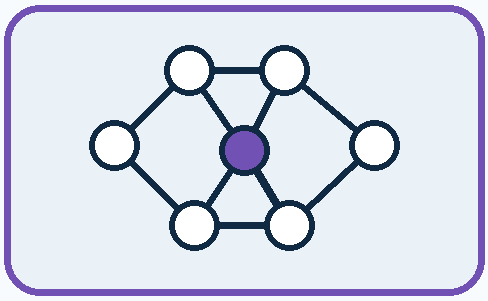
A correct flow depends on accurate catalogue entries and the declared requirements; the picture does not verify inventory.

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE A transparent sorter first checks item type, then one disambiguating feature, then sends unclear items to review.

STEP 1 — OBSERVE approved **input** fields

STEP 2 — TRACE smallest rule or learned **decision flow** with visible **branch**es

STEP 3 — CONCLUDE suggested bin, not sure, and reason

MODEL REASONING Start with a **baseline** whose **branch**es can be read aloud. Each **input** must affect a decision or be removed. Add not-sure and review paths, then explain when a learned component would earn its added complexity.

DO NOT OVERCLAIM Transparent does not mean correct; the flow still needs testing.

What new **evidence** could change this answer?

A closer explanation

Stage for 9 minutes and Quiet required No	Evidence	Next state or output
Validate	9 is a whole number in range; No is valid	Current candidate B1
Check B1	10 minutes is more than 9	Move to B2

Stage for 9 minutes and Quiet required No	Evidence	Next state or output
Check B2	8 is no more than 9; quiet is not required	Suggest B2 and stop

The current **candidate** changes from B1 to B2. The request values stay the same. After B2 passes, the **process** stops: it does not replace B2 with a later **fitting** row.

This early stop preserves the first-match rule and gives the same required suggestion as the manual **baseline**, which marked every row before selecting the first fit.

Termination needs a reason. There are only four rows, each failed check moves to the next row, and no row is revisited. A valid request therefore examines at most four **candidate**s. A success can stop sooner.

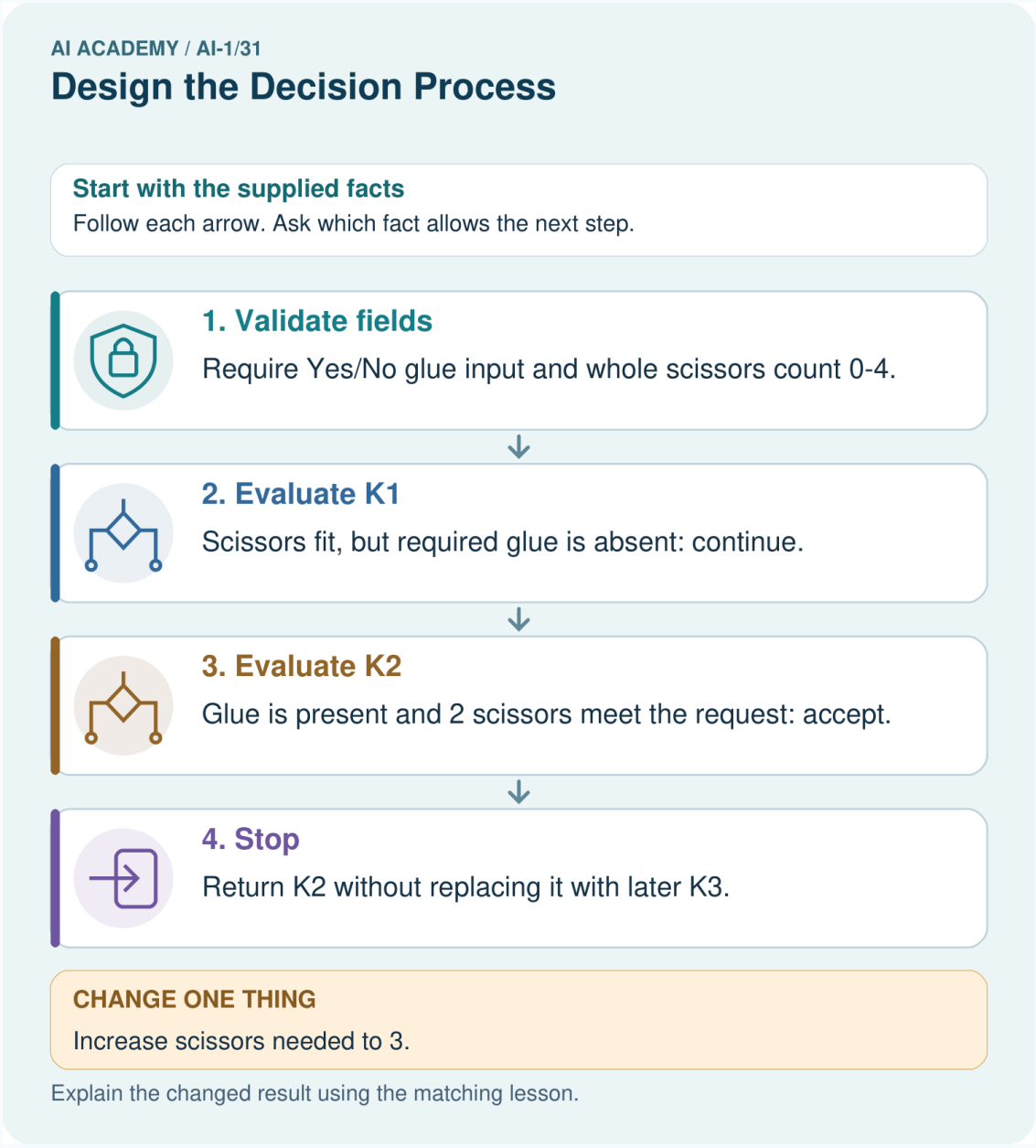
Invalid **input** stops before the search. Merely drawing an arrow around a loop is not enough; the **state** must make progress toward an end.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Design the Decision Process

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing: Increase scissors needed to 3. Predict the result before checking the **explanation**. What remains the same?

MAIN LESSON · SECTION 7

Explain the decisions the process actually made

> Keep the state, conditions, order and stops in a labelled version.

Give a reason supported by the trace

An **explanation** should identify the checks that actually justified the result. For example, B2 fits the 9-minute No request because its 8-minute duration fits and quiet was not required.

Do not add “you will love it”, which was neither an **input** nor a tested condition. For no match, report that no catalogue row met both conditions; a trace can show why.

Save the design for next week’s prototype

Save a labelled version of the **process** with its **validation**, **candidate** order, conditions, **state** changes and stopping points. A partner should be able to follow it without asking you to fill a missing step.

Next week the paper prototype will turn this design into instruction cards and a trace record. Keep any final evaluation cases reserved while developing the design.

Repair both tempting claims

- CLAIM A A longer flow must be more capable.
- CLAIM B A **learned model** does not need a visible fallback or **explanation**.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

▼ **Earlier wording, alternatives and picture transcripts**

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Picture-rule baseline	direct visible branches	easy to inspect
Learned classifier	patterns from examples	needs data evidence
Extra unused field	never changes path	remove
No fallback	forces ambiguous label	add review

Repair two tempting claims

Repair:

Repair:

A closer explanation

> **A design finds B1 compatible for 15 minutes No but keeps searching and returns B4 instead. Is B4 compatible? Does this output meet the full task requirements? Explain the difference and give the required output**

MAIN LESSON · SECTION 8

Practise: find the first broken decision

> Use concrete requests to expose premature no-match and OR/AND errors.

Challenge two incomplete rules

Consider the mistaken instruction “If B1 fails, report no match.” It would miss B2 on the 9-minute No request.

Another mistake is “If time fits or quiet fits, suggest the activity.” That could allow an activity that takes too long merely because it is quiet. Use a concrete request to show which requirement a design breaks.

Preserve the meaning of Quiet required No

A **process** must also distinguish the meaning of No from a requirement for noise. When Quiet required is No, a quiet **candidate** can still pass. For 11 minutes No, B1 fits and comes first, even though B2 also fits.

An instruction to reject all quiet activities when No is selected would change the task.

Track candidate state in W2

Use workbook W2 for the connected example “Designing a complete decision **process**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own **explanation**.

Repair the one-failed-row claim

- Use the worked case for Designing a complete decision **process**. Proposed **explanation**: “One failed row proves no match”. Decide whether this **explanation** is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to the kit stopping rule

Where does the first original rejection occur, and why is K3 not originally selected?

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Rules or learned patterns must connect **input**s to **output**s transparently.

Use a fresh example to explain **decision flow**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: A **learned model** does not need a visible fallback or **explanation**.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

> **Trace 9 minutes with Quiet required Yes through every candidate. State the reason each fails and the final output. Then explain what would happen if a failed candidate were checked again without moving to the next row. Finally, state the first step for 31 minutes with Quiet required No**

Practise with less help: Track the current candidate

Check your reasoning

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Investigate a complete flow and a premature stop

> Use the same supplied cases and contract for both flows.

The two proposed processes

Paper flow audit using the activity catalogue from this lesson. Compare two proposed flows. Flow F: validate both request fields; inspect candidates B1 through B4 in order; accept and stop only when time fits AND the quiet condition passes; otherwise advance; report no match

only after all fail. Flow G: validate, then if B1 fails report no match immediately. Predict which requests reveal a difference before tracing either flow.

Predict, trace, repair and explain termination

1. Use the same requests for both: (9 minutes,No), (5 minutes,Yes), (11 minutes,No), and (blank,Yes). Record validation, current candidate, time check, quiet check and next state/output. Keep request values unchanged during a run. Repair G's first wrong branch, trace the revised flow on the same cases and explain why at most four activity candidates are visited. Keep every original result and label these paper traces.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep the original and revised evidence

Case	Prediction	Observed result	What it means
Normal			
Edge			

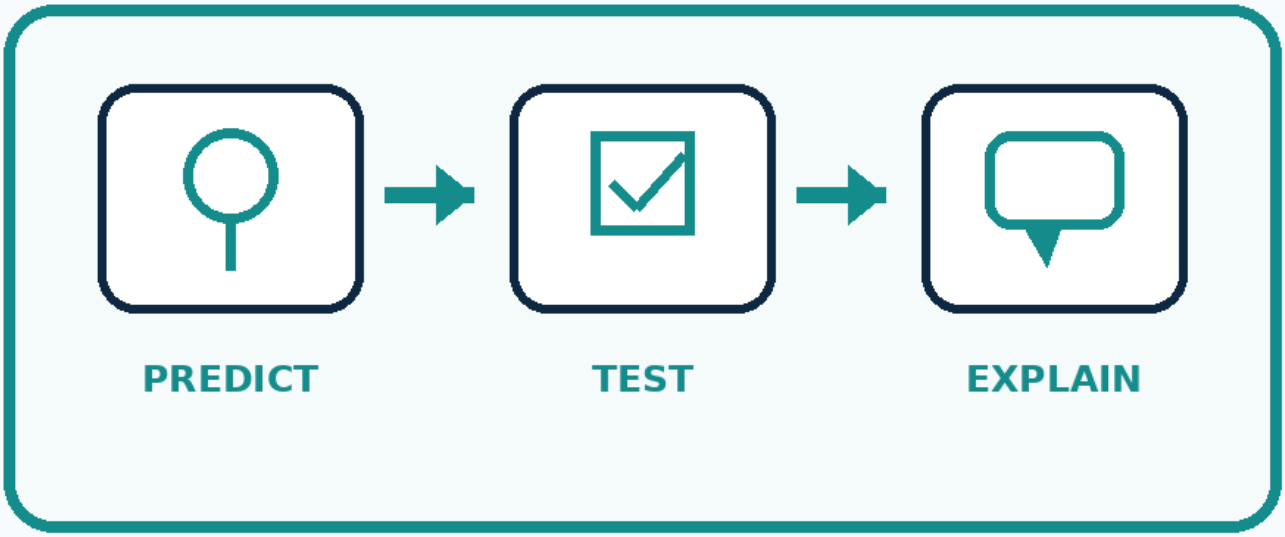
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep "not run" if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EXPLORER LAB

Smallest Useful Flow

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Create and trace the smallest useful **decision flow**.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: design the new room process

> Use C1/C2 and write validation, state changes and stops before feedback.

The independent room records and task

A new paper catalogue lists C1 first, capacity 4, quiet Yes; C2 second, capacity 8, quiet No. Valid requests give a whole group size from 1 to 10 and Quiet required Yes or No.

A room fits if capacity is at least group size and the quiet condition is met; choose the first fit. Write a complete **process** with **validation**, search, **state** change and stopping rules.

Trace 6 people No and explain the outcome for a blank quiet field. No booking takes place.

Keep your first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Design a complete room decision process

> **My independent explanation, trace and limits**

Week 31 Independent application

Close the worked **explanation**. Apply the idea to this different question without copying.

Write a decision **process**: a clear pencil goes to the pencil tray, a clear book to the shelf, and an unknown object to review. Trace a lunch box.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Complete the sorter v1 **decision flow** and annotate every **branch** with the **evidence** it uses.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Rule: if paper then P, otherwise M. **Input**s are paper, metal, glass and unknown. Repair the rule for paper and metal trays without inventing a glass tray. Trace all four **input**s.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Transfer the flow method to the sorter portfolio

> Keep the activity process and sorter rules distinct.

Connect the two contexts

The main sequence designs the paper activity recommender. Transfer the same complete-flow method to the existing sorter portfolio here; no second implementation is required. Keep each context's actual input contract, outputs and fallback rules labelled rather than copying activity-specific branches into the sorter.

This week's sorter decision

THIS WEEK'S CHECKPOINT Complete the sorter v1 **decision flow** and annotate every **branch** with the **evidence** it uses.

Draw the sorter's decision **branch**es, loop-back arrows, final **output**s and invalid-**input** path so every valid case can terminate.

Inspect every branch and uncertain case

- What real classroom need does this serve?

- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?
- When should it say not sure or ask a person?

Retain the need for testing

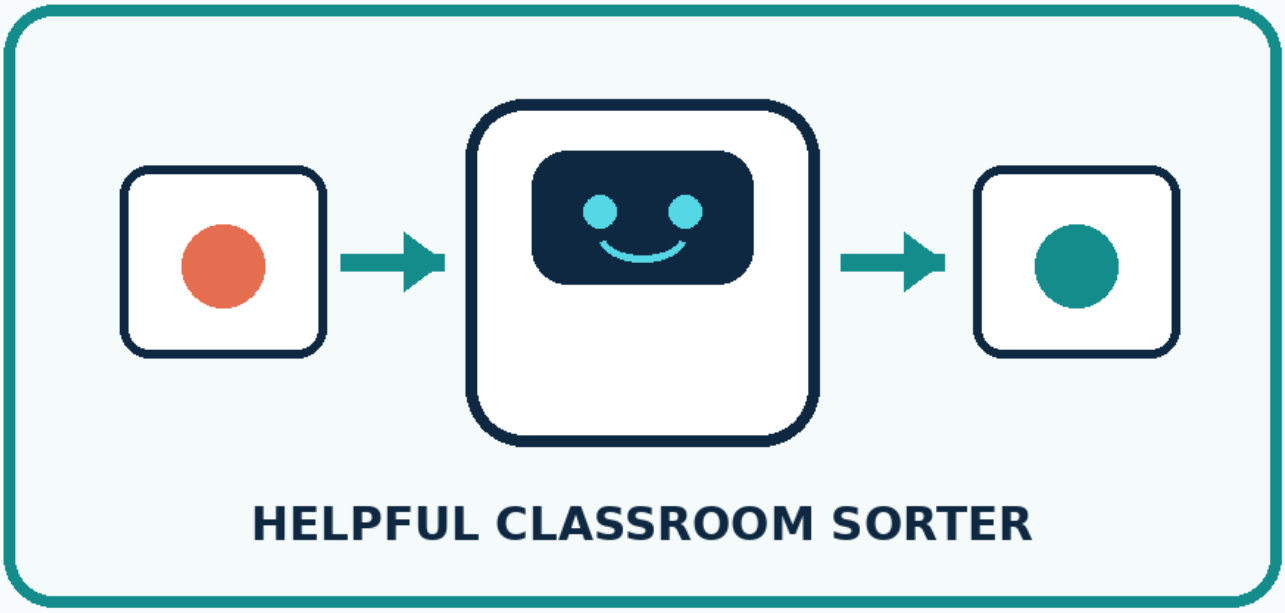
RESPONSIBLE DESIGN Transparent does not mean correct; the flow still needs testing.

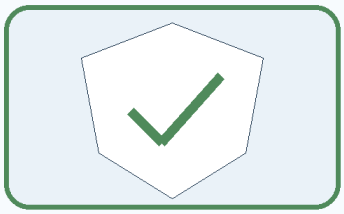
▼ **Earlier wording, alternatives and picture transcripts**

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain progress and the stopping rule

> Close your notes and trace one branch without changing the request.

Your exit explanation

- Close your notes. Explain this goal in your own words: Design a decision **process** that covers valid, invalid and no-match requests

> **My deciding step, evidence and one remaining question**

Choose the next changed-case check

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
decision flow	ordered path connecting inputs to outputs	
baseline	simple method used for comparison	
branch	path chosen by a condition	
explanation	traceable reason for an output	

Terms for the connected explanation

Word	Meaning
Validation	Checking an input against the agreed rules before using it
Candidate	An option currently being checked
State	Information that records the process's current position or values

Word	Meaning
Termination	Reaching a point where the process stops

Repair a claim using a word

- CLAIM TO REPAIR A longer flow must be more capable.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ Earlier wording, alternatives and picture transcripts

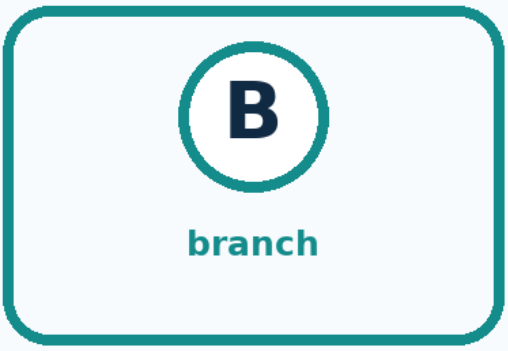
VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

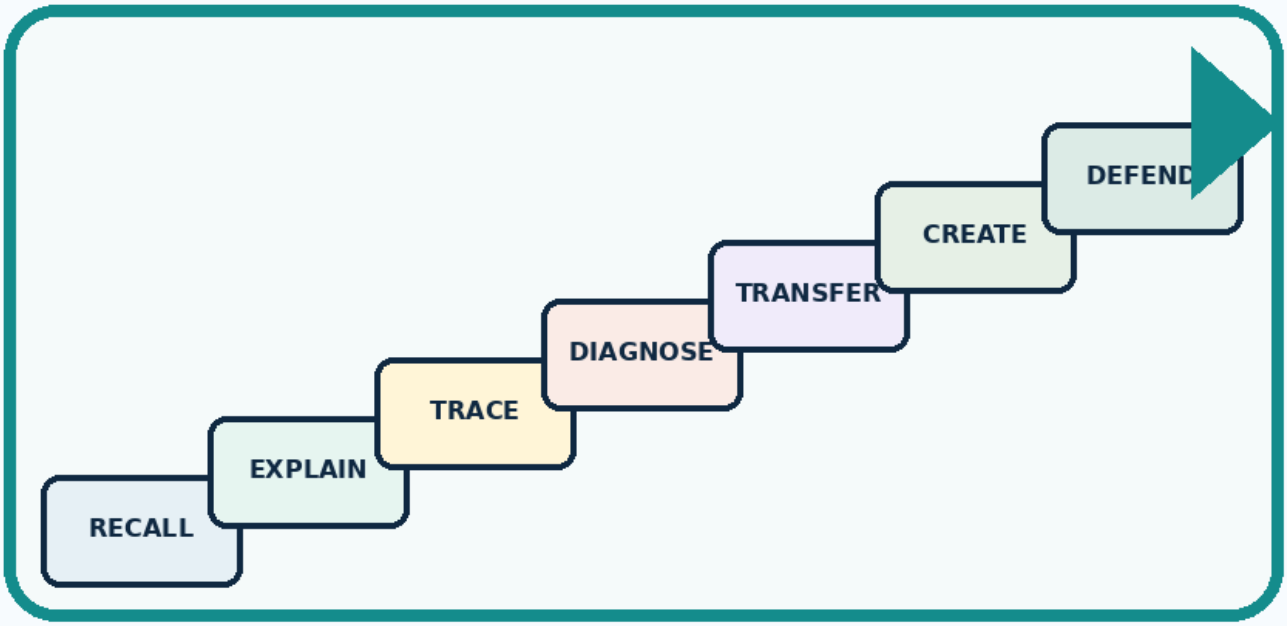
- RECALL Define **decision flow**.
- EXPLAIN Rules or learned patterns must connect **input**s to **output**s transparently.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: A longer flow must be more capable.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Design a complete room decision process is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ **State** the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ **Earlier wording, alternatives and picture transcripts**

MASTERY

Checklist for explaining this week's idea

Use after practice. Design a complete room decision process is the selected independent case.



EXPLANATION CHECK Validate the request, preserve both conditions and order, trace changing candidate state and justify all stopping paths.

Independent **evidence**:

SELF-CHECK Start with a **baseline** whose **branch**es can be read aloud. Each **input** must affect a decision or be removed. Add not-sure and review paths, then explain when a learned component would earn its added complexity.

NEXT BRIDGE Carry today's **evidence** habit into: Build a Paper Prototype.

▼ **Extra practice: Otherwise includes every remaining case**

REFERENCE · SECTION 13

Extra practice: Otherwise includes every remaining case

> Try the question before opening its worked explanation.

Read the supplied case and question

A rule sends paper to P and everything else to M. **Input**s include paper, metal and glass.

Trace glass and repair the rule.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Glass incorrectly goes to M. Add an explicit metal check and a glass or review **branch**.
Otherwise includes all remaining cases.

Evidence or stage	Value or result
Paper	Yes → P
Not paper, metal	Yes → M
Not paper, not metal	Glass or review

▼ Earlier wording, alternatives and picture transcripts

Week 31 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Otherwise includes every remaining case

A rule sends paper to P and everything else to M. **Input**s include paper, metal and glass.

Trace glass and repair the rule.

Glass incorrectly goes to M. Add an explicit metal check and a glass or review **branch**.
Otherwise includes all remaining cases.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Design a decision process that covers valid, invalid and no-match requests	Explain the deciding step, show a trace or test, and state one limit.
Trace changing state and explain why the search stops	Explain the deciding step, show a trace or test, and state one limit.
Preserve both compatibility conditions and the agreed first-match rule	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional worksheet routes craft kits by required glue and number of scissors.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional worksheet routes craft kits by required glue and number of scissors.
- Which labelled step matches this detail? Return K2 without replacing it with later K3.
- 1. Stop
- 2. Evaluate K1
- 3. Evaluate K2
- 4. Validate fields
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

After your attempt, compare with the supported **explanation**: Stop Return K2 without replacing it with later K3. A complete decision **process** specifies what to do after each result. One passing requirement is insufficient.

After a kit passes both, stopping preserves the first-match contract; automatically choosing the largest later kit would implement a different rule. A full scan with no accepted kit returns no match.

Changed-case question

- Changed-case practice: Increase scissors needed to 3. Which conclusion is supported by the supplied facts?
- 1. K3 is selected.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed case

After your attempt, compare with the supported **explanation**: K3 is selected. K1 lacks glue; K2 has only 2 scissors; K3 has glue and 4.

Repair and save your work

Review Choose **Input**s and **Output**s (ai-1/30). Goal: Turn the project outcome into precise **input** and **output** requirements. Short practice: Trace requests 10 and blank. Why should they not receive the same **explanation**?.

Return to this week and explain what you changed.

Review Humans Set Boundaries (ai-1/28). Goal: Distinguish a system's capability, the correctness of its **output** and permission to take an action. Short practice: Which original proposal is authorised?

Why do the other two need separate approval?. Return to this week and explain what you changed.

Review Generate, Retrieve, or Calculate? (ai-1/25). Goal: Distinguish retrieving a record, calculating from **input**s and generating wording. Short practice: Label each arrow with its operation.

Can the system add snacks because invitations often mention them?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

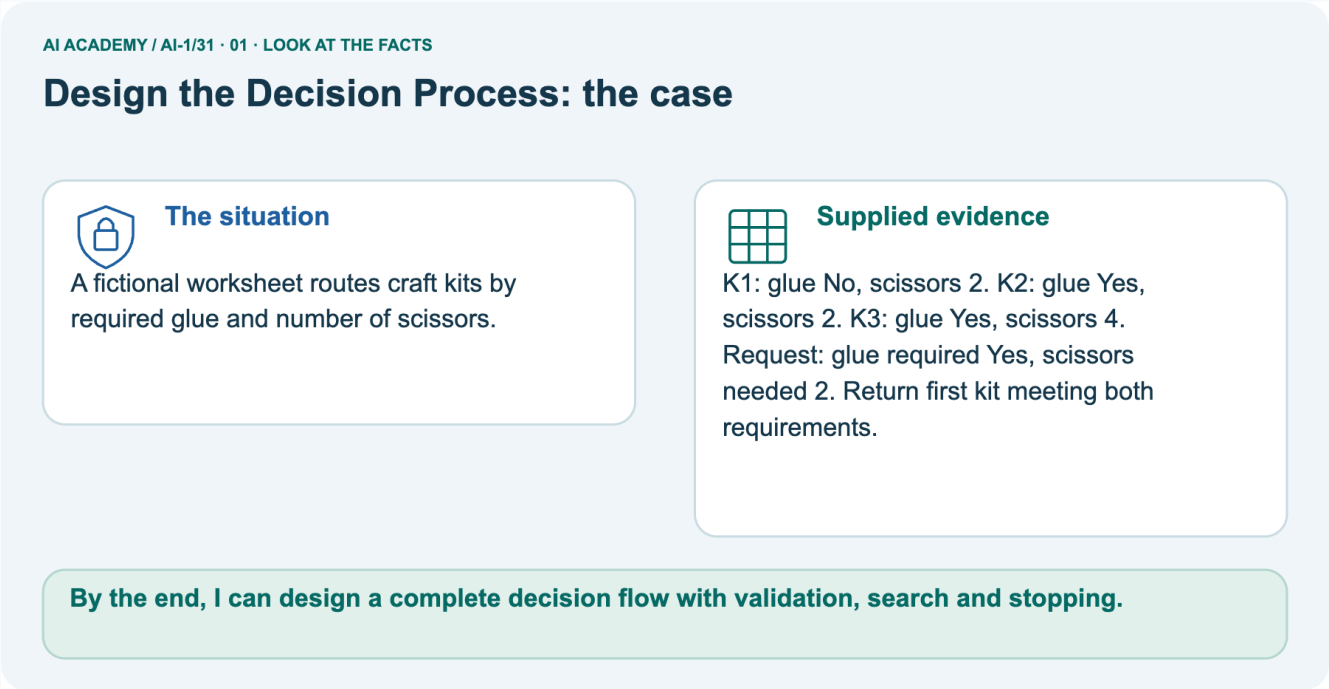
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Design the Decision Process: the case



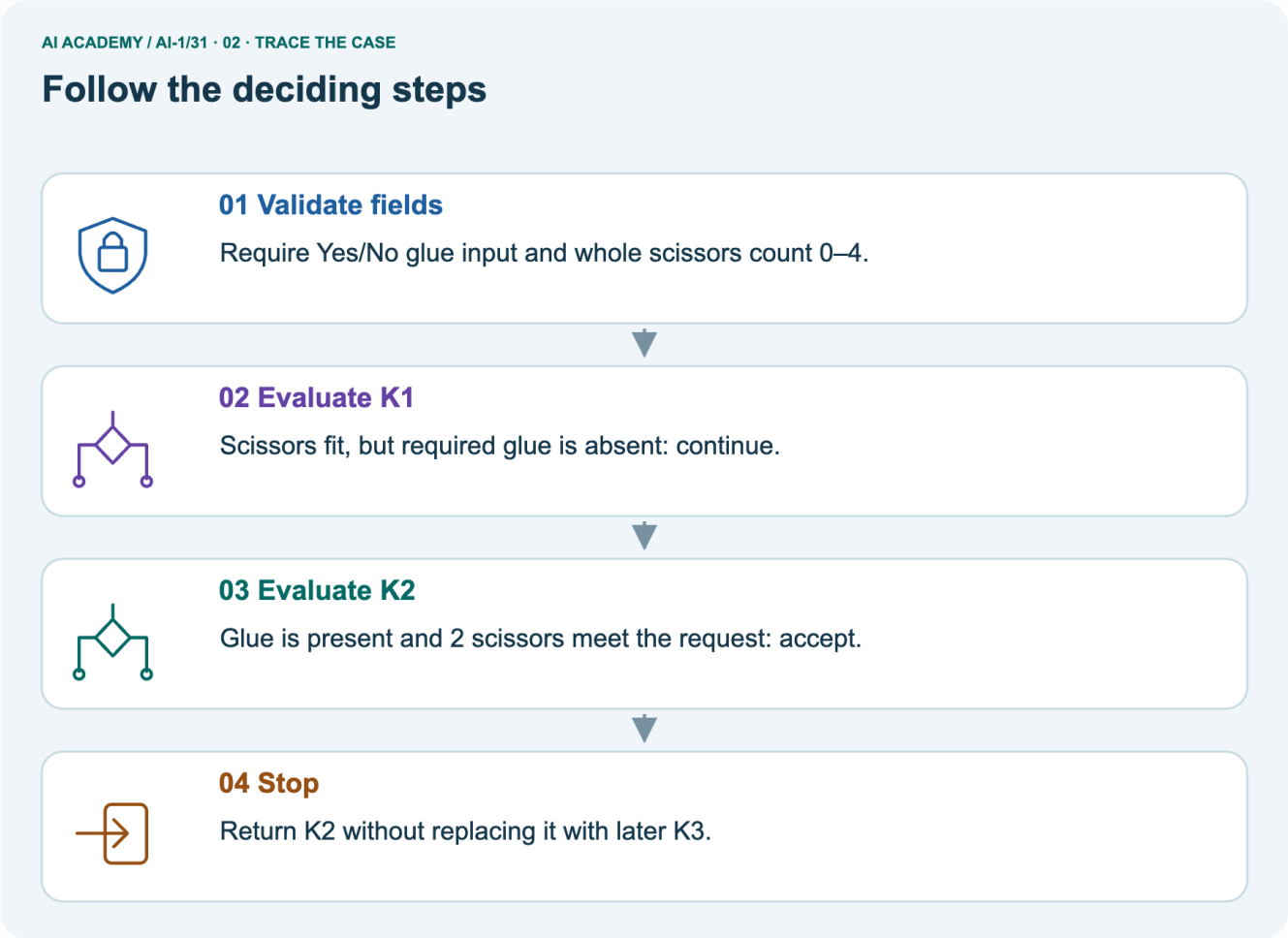
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional worksheet routes craft kits by required glue and number of scissors. Supplied facts: K1: glue No, scissors 2. K2: glue Yes, scissors 2. K3: glue Yes, scissors 4. Request: glue required Yes, scissors needed 2.

Return first kit meeting both requirements.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

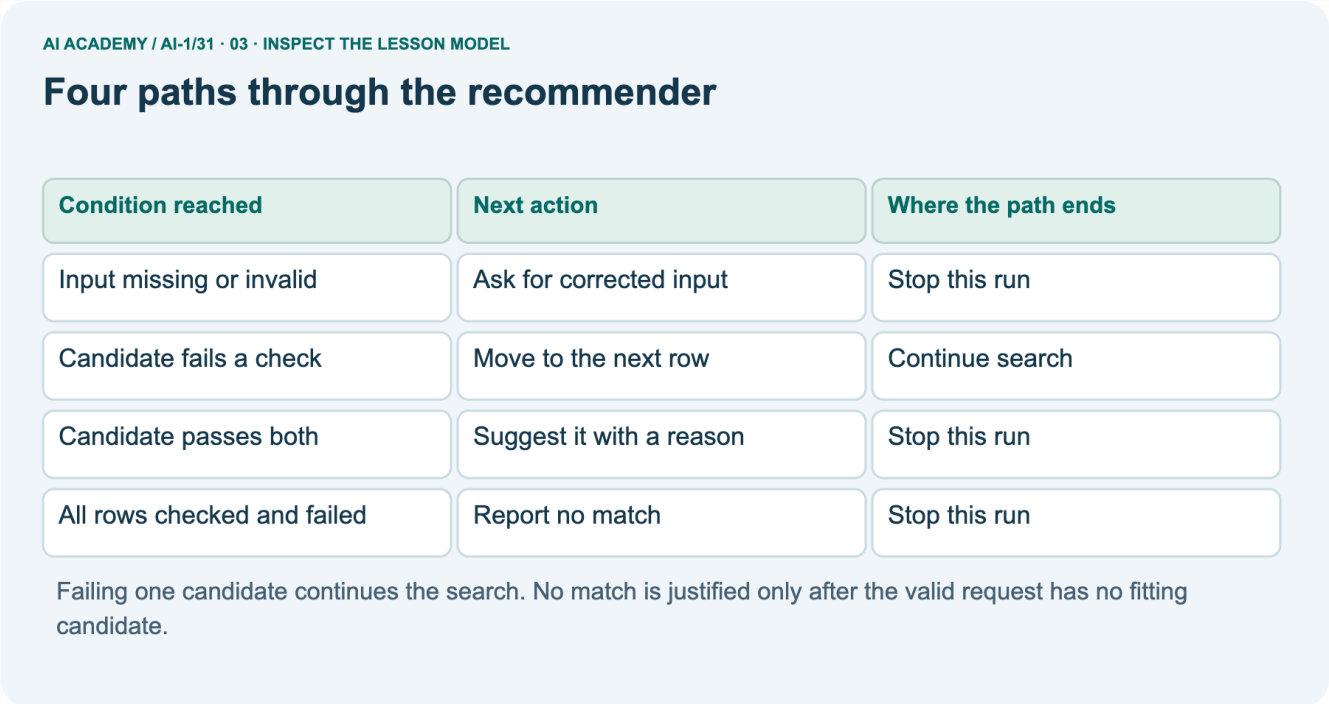


Follow the labelled picture with the matching explanation. Open the vector diagram

Validate fields: Require Yes/No glue **input** and whole scissors count 0–4. → Evaluate K1: Scissors fit, but required glue is absent: continue. → Evaluate K2: Glue is present and 2 scissors meet the request: accept. → Stop: Return K2 without replacing it with later K3.

Point to each numbered stage and explain which recorded **input** it uses.

03. Four paths through the recommender



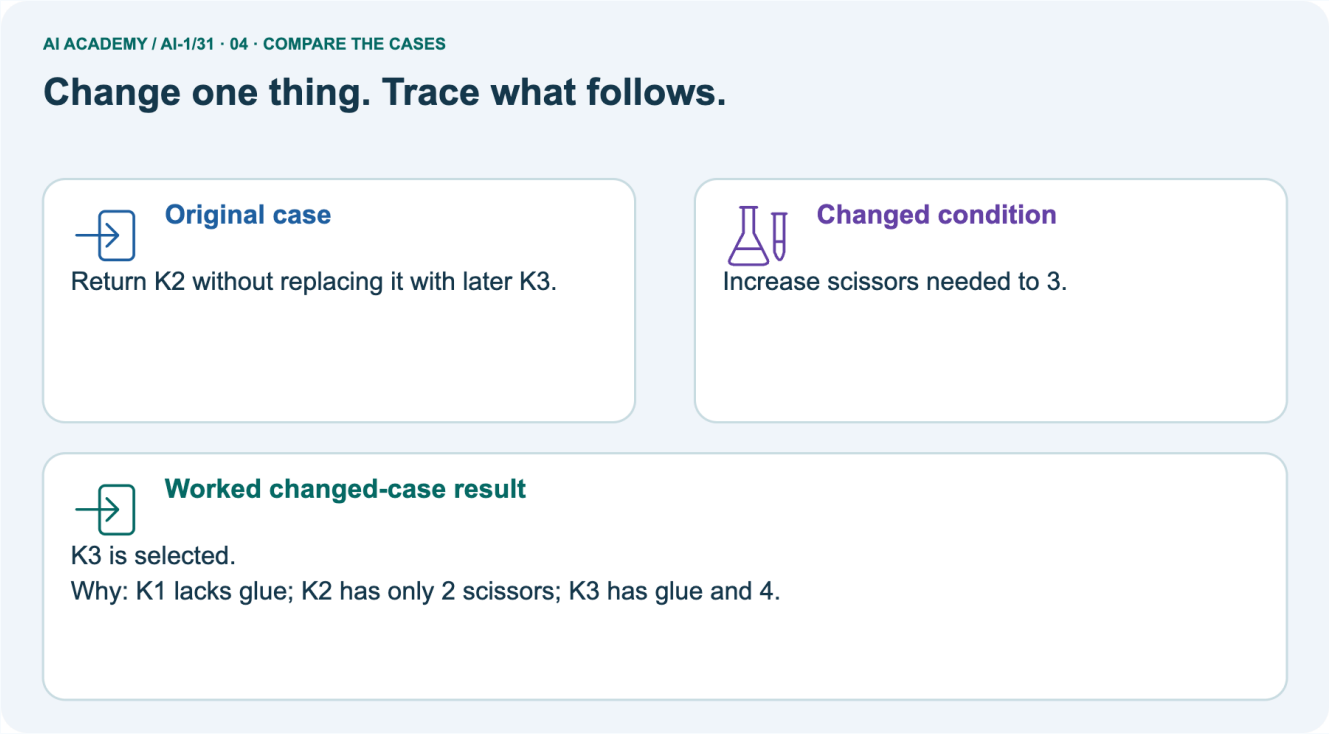
Follow the labelled picture with the matching explanation. Open the vector diagram

Four paths through the recommender. **Input** missing or invalid | Ask for corrected **input** | Stop this run. **Candidate** fails a check | Move to the next row | Continue search.

Candidate passes both | Suggest it with a reason | Stop this run. All rows checked and failed | Report no match | Stop this run

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.

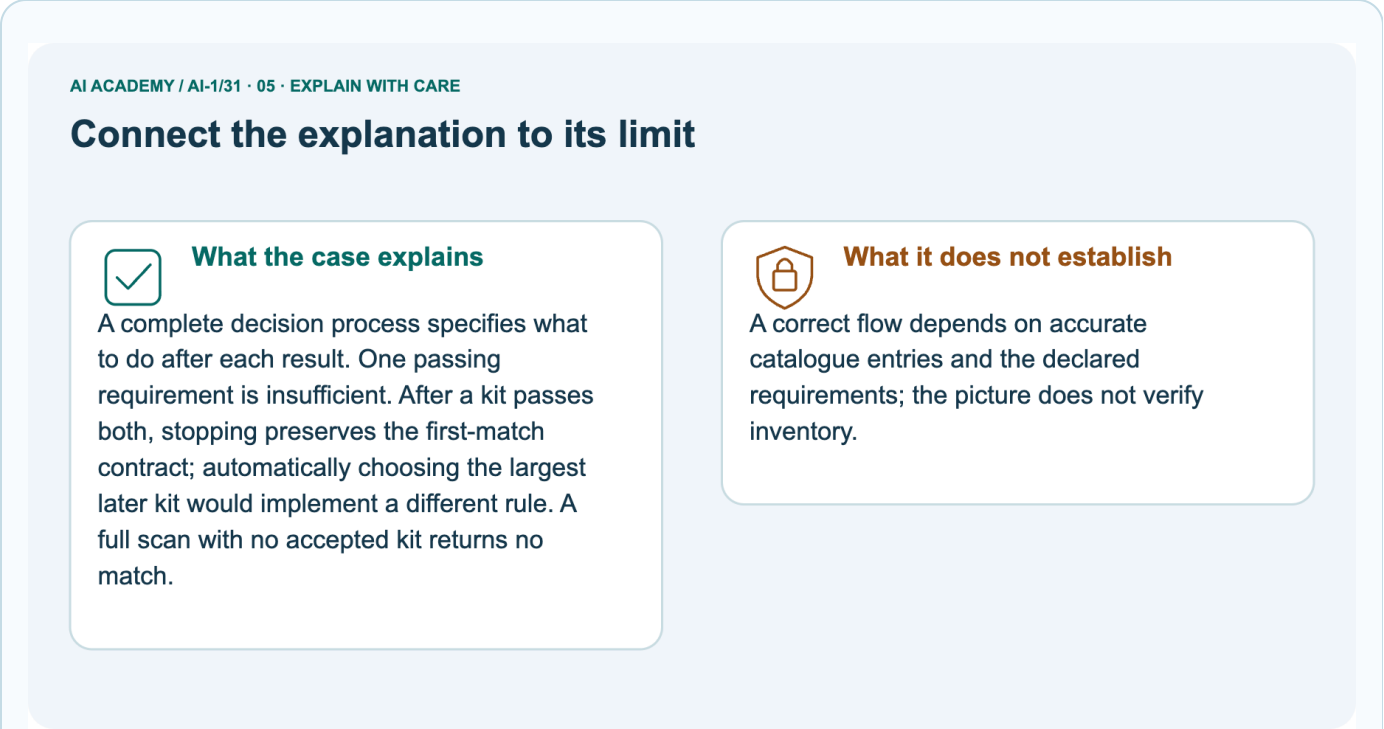


Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Return K2 without replacing it with later K3. Changed condition: Increase scissors needed to 3. Expected result: K3 is selected. Reason: K1 lacks glue; K2 has only 2 scissors; K3 has glue and 4.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

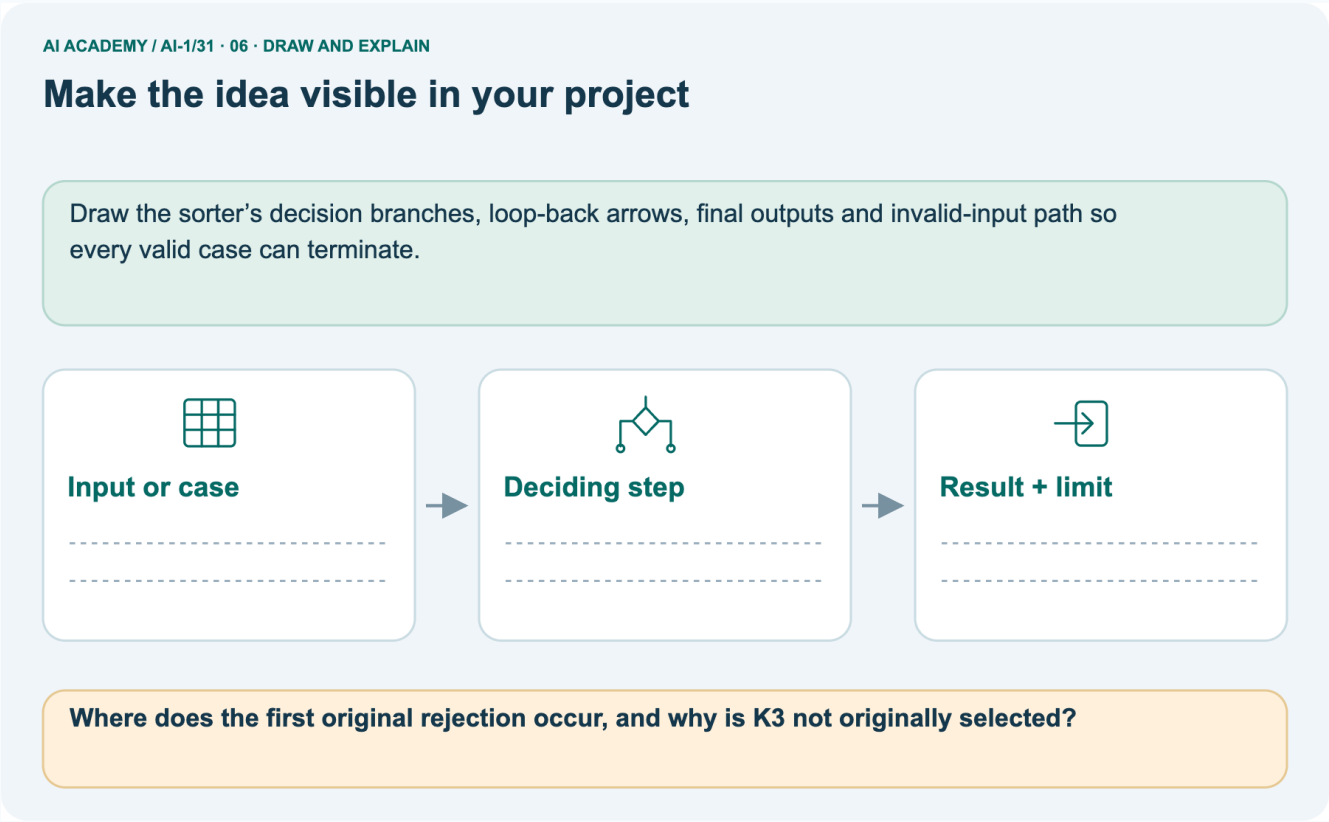
A complete decision **process** specifies what to do after each result. One passing requirement is insufficient.

After a kit passes both, stopping preserves the first-match contract; automatically choosing the largest later kit would implement a different rule. A full scan with no accepted kit returns no match.

Limit: A correct flow depends on accurate catalogue entries and the declared requirements; the picture does not verify inventory.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Draw the sorter's decision **branch**es, loop-back arrows, final **output**s and invalid-**input** path so every valid case can terminate. Draw your **input**, deciding step, result and limit.

Supported question: Where does the first original rejection occur, and why is K3 not originally selected?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Design the Decision Process: the case

02. Follow the deciding steps

03. Four paths through the recommender

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.